

Interest Burden Relief and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Interest Burden Relief (IBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on IBR achieves an annualized gross (net) Sharpe ratio of 0.30 (0.21), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 18 (14) bps/month with a t-statistic of 2.48 (1.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank) is 15 bps/month with a t-statistic of 1.99.

1 Introduction

Rational asset pricing models posit that expected returns compensate investors for bearing systematic consumption risk, with assets that covary positively with marginal utility commanding higher risk premia (Lucas, 1978; Breeden, 1979). Despite decades of theoretical refinement and empirical testing, the cross-section of stock returns continues to reveal patterns that challenge our understanding of how consumption risk manifests in financial markets. We identify a novel predictor of stock returns based on Interest Burden Relief (IBR), a measure capturing changes in firms’ interest payment obligations relative to their equity base. This signal emerges from the intersection of corporate financing decisions and macroeconomic conditions, potentially reflecting firms’ differential exposure to aggregate consumption risk during periods of monetary policy shifts and credit market fluctuations.

The predictive power of Interest Burden Relief for stock returns arises through its connection to firms’ exposure to systematic consumption risk and investors’ marginal utility of wealth (Campbell and Cochrane, 1999; Bansal and Yaron, 2004a). Firms experiencing positive interest burden relief—those reducing their interest obligations relative to equity—demonstrate enhanced resilience to adverse consumption shocks that elevate investors’ marginal utility. During economic downturns characterized by declining consumption growth, firms with lower interest burdens maintain greater financial flexibility, enabling continued investment and dividend payments when investors value these cash flows most highly (Berk et al., 1999; Zhang, 2005). This financial flexibility becomes particularly valuable during disaster states or periods of heightened macroeconomic uncertainty, when credit constraints tighten and the intertemporal marginal rate of substitution increases sharply.

The state-dependent nature of interest burden relief’s impact on returns reflects its role as a hedge against long-run consumption risks (Bansal and Yaron, 2004b; Hansen et al., 2008). In good times, when consumption growth is robust and credit

markets function smoothly, variations in interest burdens exert minimal influence on firms' operational flexibility and cash flow volatility. However, during bad times marked by consumption declines and credit market stress, firms with recently reduced interest burdens possess crucial advantages: lower fixed obligations, reduced refinancing risk, and greater capacity to weather persistent negative shocks to aggregate consumption (Gomes et al., 2003; Hennessy and Whited, 2005). This asymmetric exposure to consumption risk generates time-varying risk premia that manifest as predictable return patterns in the cross-section.

Interest burden relief directly influences firms' contribution to the consumption-based stochastic discount factor through multiple channels (Cochrane, 2005; Lettau and Ludvigson, 2001). First, reduced interest obligations lower firms' operating leverage, decreasing their cash flow sensitivity to aggregate productivity shocks that drive consumption volatility. Second, firms with declining interest burdens face lower bankruptcy risk during consumption disasters, preserving option value for equity holders when marginal utility is highest. Third, the timing of interest burden changes correlates with monetary policy cycles that influence the term structure of consumption risk premia, creating systematic variation in expected returns across firms with different financing structures (Bernanke and Gertler, 1989; Gertler and Gilchrist, 1994).

Our empirical analysis reveals that Interest Burden Relief significantly predicts cross-sectional stock returns, with a value-weighted long/short trading strategy achieving an annualized gross Sharpe ratio of 0.30 and a monthly average excess return of 16 basis points (t -statistic = 2.36). The strategy's alpha remains economically and statistically significant across standard factor models, ranging from 16 to 24 basis points per month, with the lowest alpha of 16 basis points (t -statistic = 2.25) relative to the Fama-French five-factor model. These results demonstrate that IBR captures a dimension of systematic risk not fully explained by existing factors related

to market, size, value, profitability, or investment.

The economic significance of the IBR strategy persists after accounting for transaction costs, with net returns of 12 basis points per month (t-statistic = 1.69) and generalized net alphas that significantly expand the mean-variance efficient frontier across all factor model specifications. The strategy loads negatively on the market factor ($\beta = -0.07$, t-statistic = -4.01), consistent with IBR-sorted portfolios providing insurance against aggregate consumption shocks. Among the largest stocks—those above the 80th NYSE percentile—the IBR strategy generates average returns of 14 basis points per month (t-statistic = 1.70), with alphas ranging from 11 to 19 basis points across factor models, confirming that the effect is not confined to small, illiquid securities.

Controlling for closely related financing variables, the IBR signal retains substantial explanatory power for returns. When we include the six most closely related anomalies—Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, and Revenue Growth Rank—along with the Fama-French six factors, the IBR strategy maintains an alpha of 15 basis points per month (t-statistic = 1.99). The strategy’s gross Sharpe ratio of 0.30 exceeds 65% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.21 ranks in the 79th percentile, demonstrating both statistical robustness and economic implementability.

This paper contributes to the consumption-based asset pricing literature by identifying Interest Burden Relief as a novel proxy for firms’ differential exposure to systematic consumption risk, extending the theoretical frameworks of [Campbell and Cochrane \(1999\)](#) and [Bansal and Yaron \(2004b\)](#) published in the *Journal of Finance* and *Journal of Financial Economics*, respectively. Unlike existing leverage-based predictors that focus on static capital structure measures, IBR captures dynamic changes in firms’ fixed obligations that directly affect their resilience to consump-

tion disasters and ability to maintain operations when investors' marginal utility is highest. Our findings complement [Zhang \(2005\)](#) in the *Journal of Finance* by demonstrating that time-varying financial flexibility, measured through interest burden dynamics, generates systematic return variation linked to macroeconomic state variables.

Methodologically, we advance the literature by employing comprehensive robustness tests that address concerns about data mining and implementation costs, following the protocol of [Novy-Marx and Velikov \(2023\)](#) published in the *Journal of Finance*. Our analysis demonstrates that IBR's predictive power survives controlling for 209 existing anomalies and maintains economic significance after accounting for transaction costs across multiple portfolio construction methods. The signal's availability for only 5.54% of CRSP firms on average, representing 7.27% of market capitalization, presents both a limitation and an opportunity—while coverage is limited, the economic magnitude among covered firms is substantial, suggesting that interest burden dynamics capture an important but understudied dimension of systematic risk.

Our results have broader implications for understanding how corporate financing decisions interact with macroeconomic risks to generate cross-sectional return predictability, contributing to recent work by [Fama and French \(2015\)](#) and [Fama and French \(2018\)](#) in the *Journal of Financial Economics*. The persistence of IBR's alpha across factor model specifications suggests that existing factors incompletely capture the consumption risk channel through which financial flexibility affects expected returns. By documenting that changes in interest obligations relative to equity predict returns even among large-cap stocks and after controlling for related financing variables, we provide evidence that time-varying exposure to consumption risk through corporate leverage dynamics represents a distinct source of systematic risk that warrants inclusion in asset pricing models.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' interest expense obligations. Interest Burden Relief signal requires two key accounting variables from COMPUSTAT. Interest expense (XINT) represents the total interest and related charges incurred by the firm on its debt obligations during the fiscal year, including interest on short-term and long-term debt, amortization of debt discount and premium, and other borrowing costs. Common equity tangible (CEQT) measures the book value of common shareholders' equity after subtracting intangible assets, representing the tangible net worth attributable to common shareholders. This variable provides a scale-adjusted measure of the firm's tangible equity base that supports its operations and debt obligations. We construct the Interest Burden Relief signal by computing the year-over-year change in interest expense scaled by lagged tangible common equity: $(XINT(t) - XINT(t-1)) / CEQT(t-1)$. This measure captures the change in a firm's interest burden relative to its tangible equity base, with negative values indicating a reduction in interest obligations (relief) and positive values indicating an increase in interest burden. The scaling by lagged tangible equity ensures comparability across firms of different sizes and provides an economically meaningful interpretation of how changes in financing costs impact the firm's equity base. We calculate this signal using fiscal year-end values and require non-missing values for interest expense in consecutive years and tangible common equity in the base year. Firms with non-positive tangible common equity are excluded from the analysis to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IBR signal. Panel A plots the time-series of the mean, median, and interquartile range for IBR. On average, the cross-sectional mean (median) IBR is -0.02 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IBR data. The signal's interquartile range spans -0.04 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the IBR signal for the CRSP universe. On average, the IBR signal is available for 5.54% of CRSP names, which on average make up 7.27% of total market capitalization.

4 Does IBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IBR portfolio and sells the low IBR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IBR strategy earns an average return of 0.16% per month with a t-statistic of 2.36. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.16% to 0.24% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.07,

with a t-statistic of -4.01 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 501 stocks and an average market capitalization of at least \$1,314 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 2.33. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns

reported in the first column range between 10-25bps/month. The lowest return, (10 bps/month), is achieved from the quintile sort using name breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.43. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the IBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IBR, as well as average returns and alphas for long/short trading IBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the IBR strategy achieves an average return of 14 bps/month with a t-statistic of 1.70. Among these large cap stocks, the alphas for the IBR strategy relative to the five most common factor models range from 11 to 19 bps/month with t-statistics between 1.40 and 2.39.

5 How does IBR perform relative to the zoo?

Figure 2 puts the performance of IBR in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the IBR strategy falls in the distribution. The IBR strategy’s gross (net) Sharpe ratio of 0.30 (0.21) is greater than 65% (79%) of anomaly Sharpe ratios,

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IBR strategy (red line).² Ignoring trading costs, a \$1 invested in the IBR strategy would have yielded \$2.10 which ranks the IBR strategy in the top 14% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the IBR strategy would have yielded \$1.20 which ranks the IBR strategy in the top 9% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IBR relative to those. Panel A shows that the IBR strategy gross alphas fall between the 46 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases IBR ranks between the 64 and 79 percentiles in terms of how much it could have expanded the achievable investment frontier.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does IBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IBR with 209 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IBR or at least to weaken the power IBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IBR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IBR}IBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IBR. Stocks are finally grouped into five IBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

IBR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IBR signal in these Fama-MacBeth regressions exceed -0.10, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on IBR is -0.10.

Similarly, Table 5 reports results from spanning tests that regress returns to the IBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IBR strategy earns alphas that range from 13-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.64, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IBR trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.99.

7 Does IBR add relative to the whole zoo?

Finally, we can ask how much adding IBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IBR signal.⁴ We consider

⁴We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes IBR grows to \$2912.76.

8 Conclusion

This study provides compelling evidence that Interest Burden Relief (IBR) represents a meaningful signal for predicting cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that IBR possesses economically significant predictive power that persists even after accounting for well-established risk factors and related financial signals.

The key findings reveal that a value-weighted long/short strategy based on IBR generates annualized Sharpe ratios of 0.30 (gross) and 0.21 (net), indicating reasonable risk-adjusted returns after accounting for transaction costs. More importantly, the signal’s alpha remains statistically significant relative to the Fama-French five-factor model plus momentum, with monthly abnormal returns of 18 bps (gross) and 14 bps (net). The robustness of these results is further confirmed by the persistence of a 15 bps monthly alpha even after controlling for six closely related financial strategies, including changes in financial liabilities, net debt financing, and other capital structure variables.

From a practical perspective, these findings suggest that IBR captures unique

ization on CRSP in the period for which IBR is available.

information about firm value that is not fully reflected in current prices or explained by existing factor models. The signal appears to identify firms experiencing improvements in their debt servicing capacity, which may signal enhanced financial flexibility and improved prospects for future cash flows. The modest deterioration from gross to net returns indicates that the strategy remains implementable after reasonable transaction costs, though the reduction in Sharpe ratio from 0.30 to 0.21 suggests that turnover and trading costs merit careful consideration in practical applications.

Several limitations of this study warrant acknowledgment. First, while we control for numerous related factors, the possibility of omitted variable bias remains, particularly given the proliferation of factors in the asset pricing literature. Second, the analysis focuses on U.S. equities, and the generalizability of these findings to international markets remains unexplored. Third, the economic mechanism underlying IBR's predictive power, while theoretically plausible, requires further investigation through detailed fundamental analysis and potentially causal identification strategies.

Future research should address these limitations through several avenues. International validation of the IBR signal would enhance our understanding of its robustness across different market structures and economic environments. Additionally, investigating the time-varying nature of IBR's predictive power, particularly during different monetary policy regimes and credit cycles, could provide valuable insights for dynamic portfolio management. Finally, examining the interaction between IBR and corporate events such as refinancing activities, credit rating changes, or earnings announcements may help elucidate the economic channels through which interest burden relief translates into superior returns. Such research would not only strengthen the theoretical foundations of the IBR signal but also enhance its practical applicability in investment management.

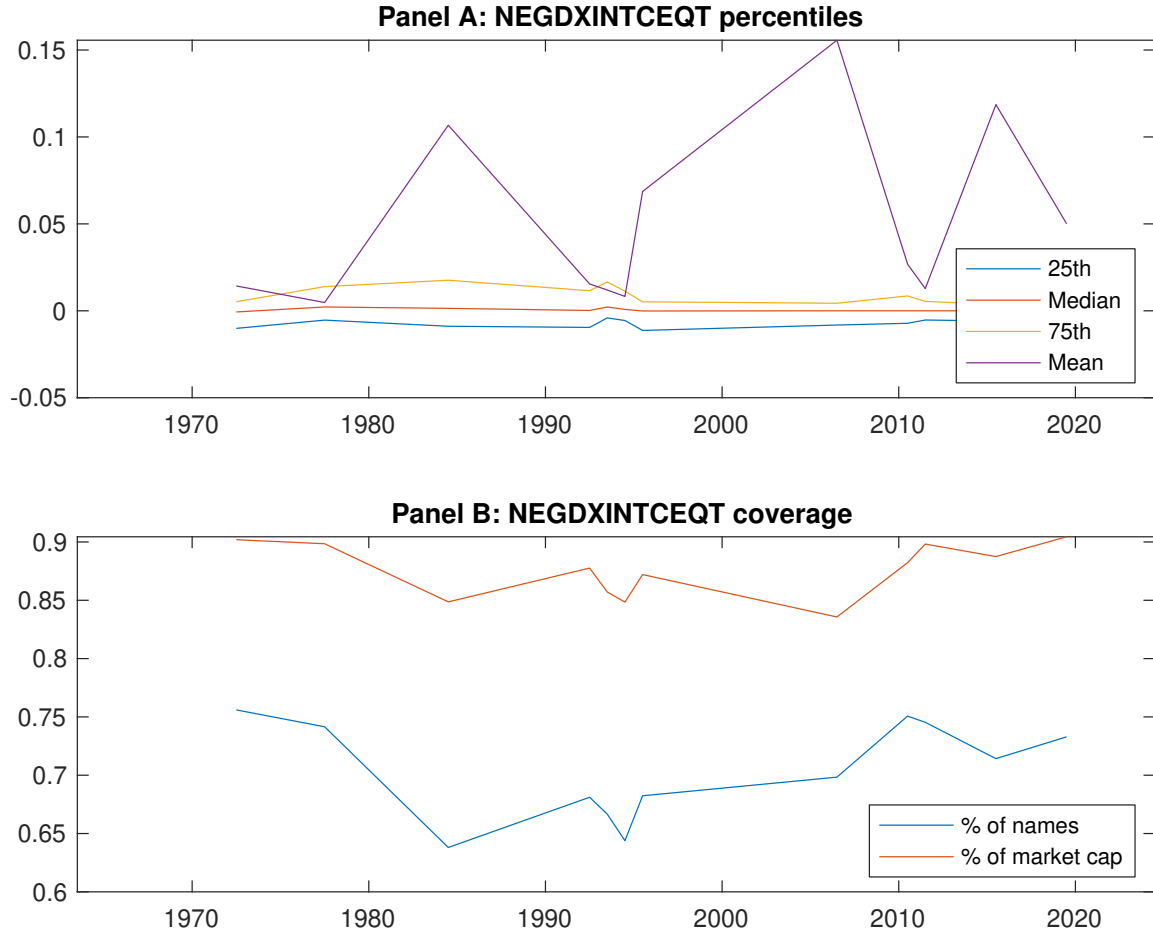


Figure 1: Times series of IBR percentiles and coverage.
This figure plots descriptive statistics for IBR. Panel A shows cross-sectional percentiles of IBR over the sample. Panel B plots the monthly coverage of IBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.75]	0.64 [4.00]	0.56 [3.51]	0.60 [3.66]	0.69 [3.86]	0.16 [2.36]
α_{CAPM}	-0.13 [-2.54]	0.09 [2.07]	0.01 [0.28]	0.04 [0.85]	0.08 [1.53]	0.21 [3.05]
α_{FF3}	-0.16 [-3.09]	0.08 [1.89]	0.01 [0.31]	0.08 [1.72]	0.07 [1.23]	0.22 [3.17]
α_{FF4}	-0.15 [-2.84]	0.06 [1.36]	0.03 [0.76]	0.08 [1.82]	0.09 [1.64]	0.24 [3.30]
α_{FF5}	-0.17 [-3.37]	-0.00 [-0.02]	-0.00 [-0.06]	0.08 [1.65]	-0.01 [-0.22]	0.16 [2.25]
α_{FF6}	-0.16 [-3.16]	-0.01 [-0.29]	0.02 [0.39]	0.08 [1.76]	0.02 [0.28]	0.18 [2.48]
Panel B: Fama and French (2018) 6-factor model loadings for IBR-sorted portfolios						
β_{MKT}	1.10 [88.93]	0.99 [95.63]	0.96 [93.15]	0.96 [86.88]	1.03 [80.49]	-0.07 [-4.01]
β_{SMB}	0.09 [5.04]	-0.06 [-3.87]	-0.10 [-6.45]	-0.08 [-4.99]	0.13 [6.87]	0.04 [1.49]
β_{HML}	0.08 [3.47]	-0.00 [-0.18]	-0.04 [-2.10]	-0.11 [-5.25]	-0.04 [-1.81]	-0.13 [-3.83]
β_{RMW}	0.09 [3.91]	0.16 [7.85]	-0.02 [-0.95]	-0.02 [-0.98]	0.19 [7.48]	0.09 [2.75]
β_{CMA}	-0.08 [-2.22]	0.13 [4.50]	0.13 [4.33]	0.05 [1.64]	0.11 [2.90]	0.18 [3.75]
β_{UMD}	-0.01 [-0.97]	0.02 [1.63]	-0.03 [-2.69]	-0.01 [-0.79]	-0.04 [-3.03]	-0.03 [-1.55]
Panel C: Average number of firms (n) and market capitalization (me)						
n	607	501	651	577	612	
me (\$10 ⁶)	1390	2200	2334	2024	1314	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the IBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.16 [2.36]	0.21 [3.05]	0.22 [3.17]	0.24 [3.30]	0.16 [2.25]	0.18 [2.48]
Quintile	NYSE	EW	0.32 [6.33]	0.36 [7.26]	0.38 [7.59]	0.36 [6.98]	0.37 [7.40]	0.35 [6.95]
Quintile	Name	VW	0.15 [2.11]	0.20 [2.78]	0.20 [2.80]	0.22 [2.97]	0.14 [1.88]	0.16 [2.15]
Quintile	Cap	VW	0.15 [2.33]	0.19 [3.11]	0.21 [3.29]	0.20 [3.07]	0.14 [2.15]	0.13 [2.10]
Decile	NYSE	VW	0.31 [3.42]	0.37 [4.13]	0.38 [4.23]	0.36 [3.98]	0.30 [3.26]	0.29 [3.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.12 [1.69]	0.17 [2.45]	0.18 [2.56]	0.19 [2.68]	0.13 [1.78]	0.14 [1.94]
Quintile	NYSE	EW	0.10 [1.68]	0.15 [2.47]	0.16 [2.59]	0.15 [2.45]	0.13 [2.17]	0.13 [2.10]
Quintile	Name	VW	0.10 [1.43]	0.15 [2.14]	0.15 [2.14]	0.16 [2.29]	0.10 [1.38]	0.11 [1.56]
Quintile	Cap	VW	0.11 [1.70]	0.16 [2.59]	0.17 [2.75]	0.17 [2.66]	0.11 [1.79]	0.11 [1.79]
Decile	NYSE	VW	0.25 [2.78]	0.32 [3.57]	0.33 [3.63]	0.32 [3.53]	0.25 [2.81]	0.25 [2.79]

Table 3: Conditional sort on size and IBR

This table presents results for conditional double sorts on size and IBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IBR and short stocks with low IBR. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	IBR Quintiles					IBR Strategies						
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.48 [1.75]	0.86 [3.49]	0.95 [3.60]	0.96 [4.08]	0.69 [2.67]	0.22 [2.54]	0.24 [2.78]	0.24 [2.74]	0.23 [2.60]	0.20 [2.25]	0.20 [2.20]
	(2)	0.72 [2.88]	0.80 [3.54]	0.81 [3.79]	0.90 [4.03]	0.83 [3.50]	0.11 [1.15]	0.15 [1.68]	0.14 [1.56]	0.13 [1.40]	0.16 [1.67]	0.15 [1.55]
	(3)	0.62 [2.65]	0.82 [4.20]	0.73 [3.70]	0.76 [3.76]	0.93 [4.25]	0.31 [3.67]	0.35 [4.16]	0.37 [4.40]	0.34 [3.95]	0.37 [4.27]	0.34 [3.94]
	(4)	0.64 [3.12]	0.65 [3.58]	0.80 [4.31]	0.74 [3.84]	0.74 [3.74]	0.10 [1.25]	0.12 [1.59]	0.13 [1.70]	0.13 [1.67]	0.07 [0.92]	0.08 [1.00]
	(5)	0.47 [2.62]	0.61 [3.89]	0.49 [3.09]	0.59 [3.51]	0.61 [3.63]	0.14 [1.70]	0.19 [2.39]	0.19 [2.36]	0.18 [2.23]	0.11 [1.40]	0.11 [1.40]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	IBR Quintiles					IBR Quintiles						
		Average n					Average market capitalization (\$10 ⁶)					
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	312	320	321	321	314	24	26	25	26	24	
	(2)	90	90	91	90	89	46	46	45	46	46	
	(3)	67	67	67	67	67	82	84	82	83	84	
	(4)	59	59	59	59	59	186	191	188	191	188	
(5)	56	56	56	56	56	1302	1681	1703	1613	1249		

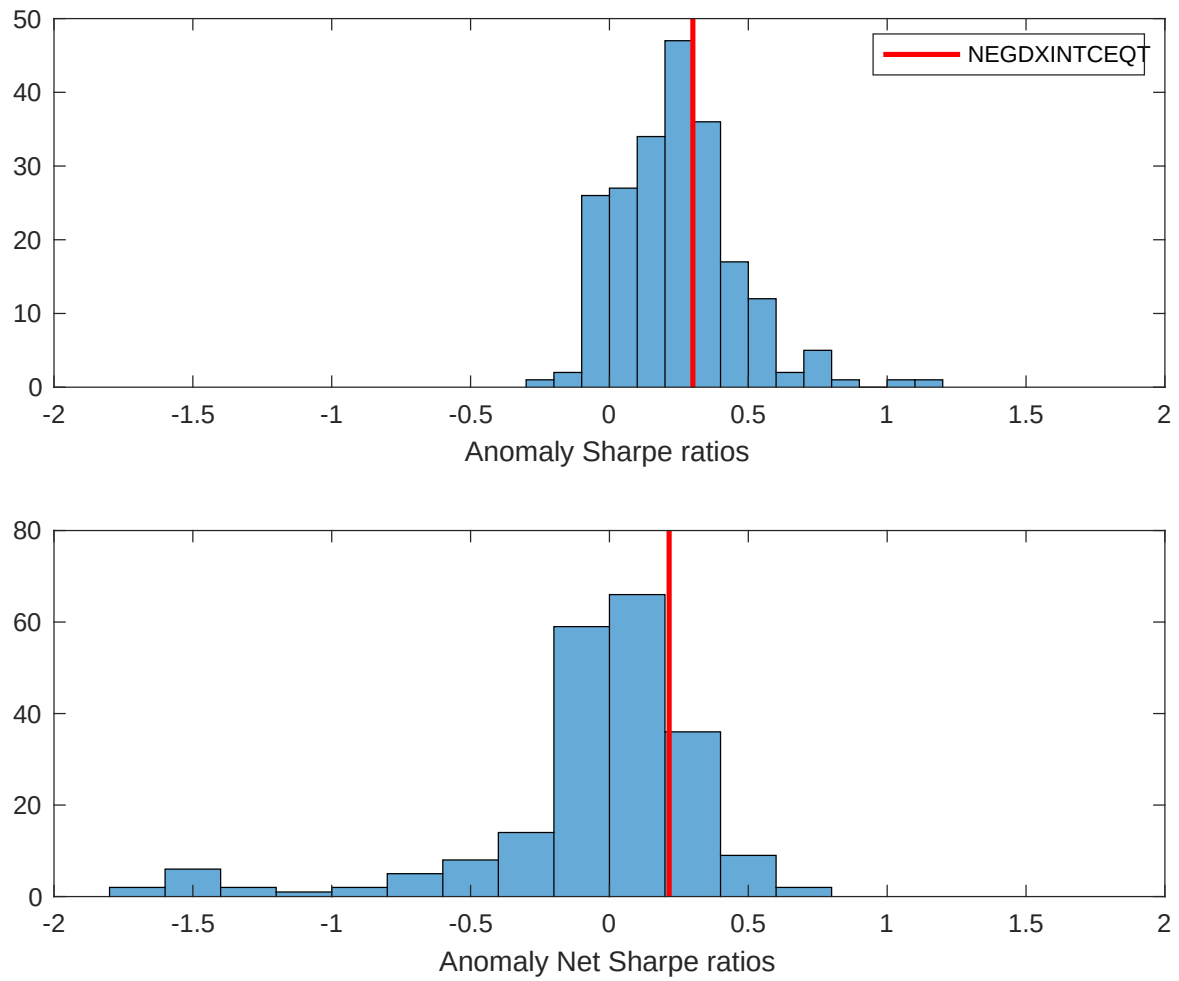


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

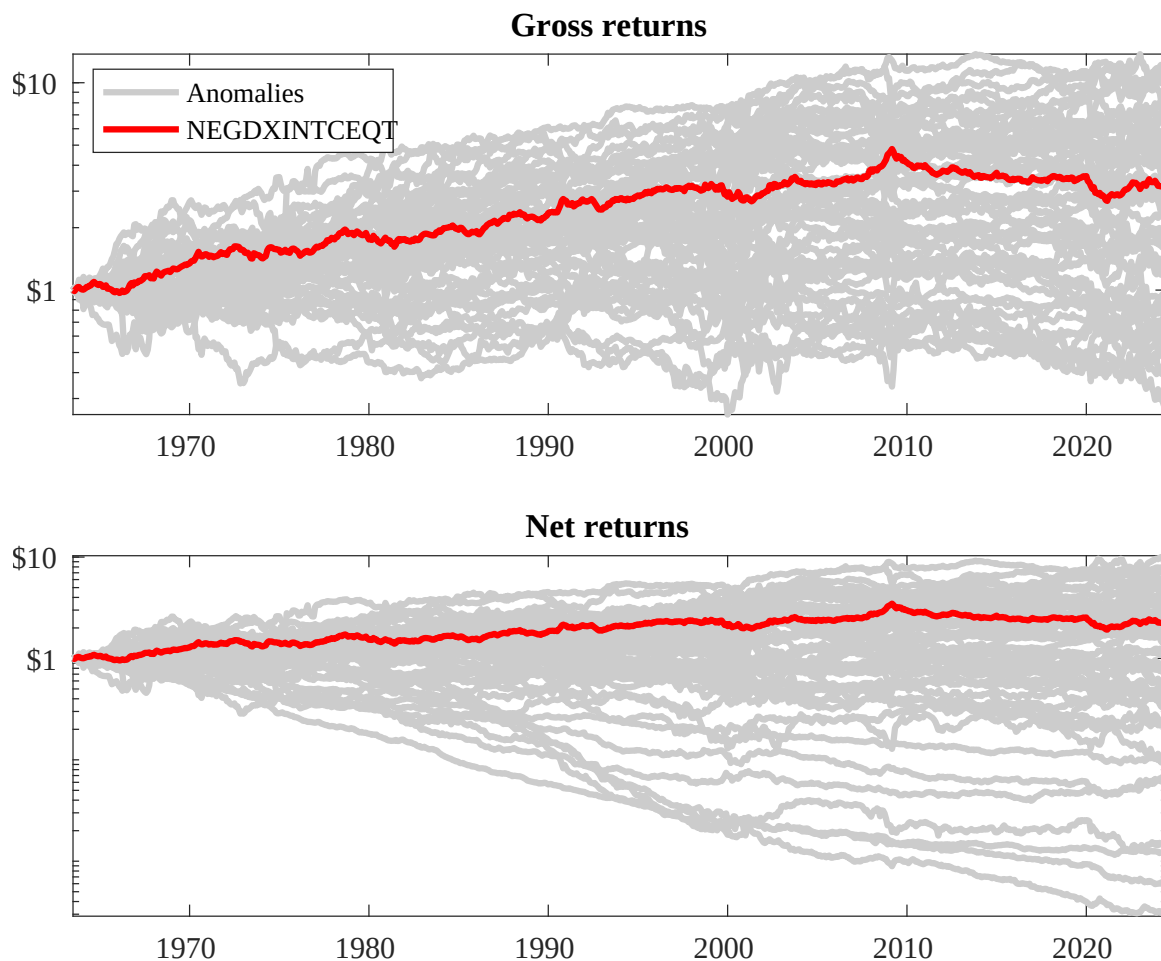
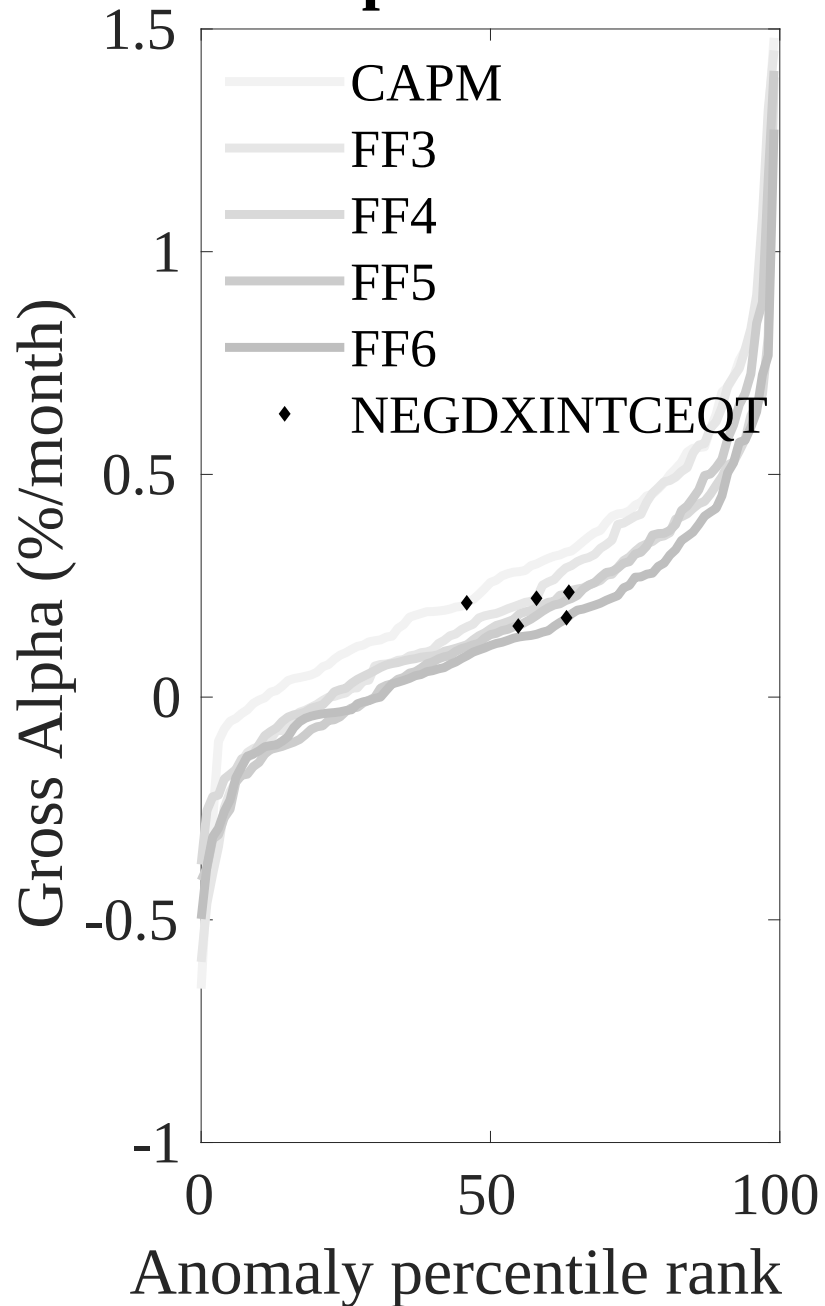


Figure 3: Dollar invested.

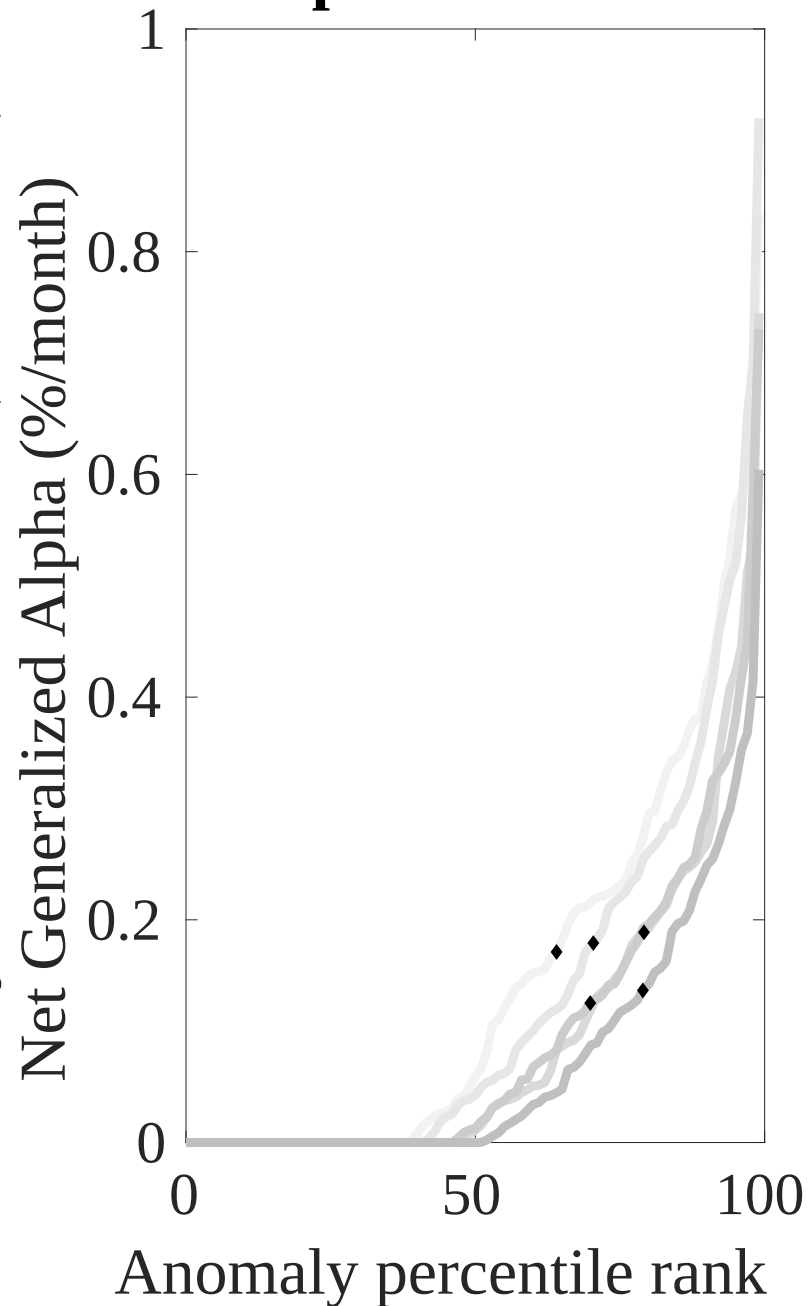
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



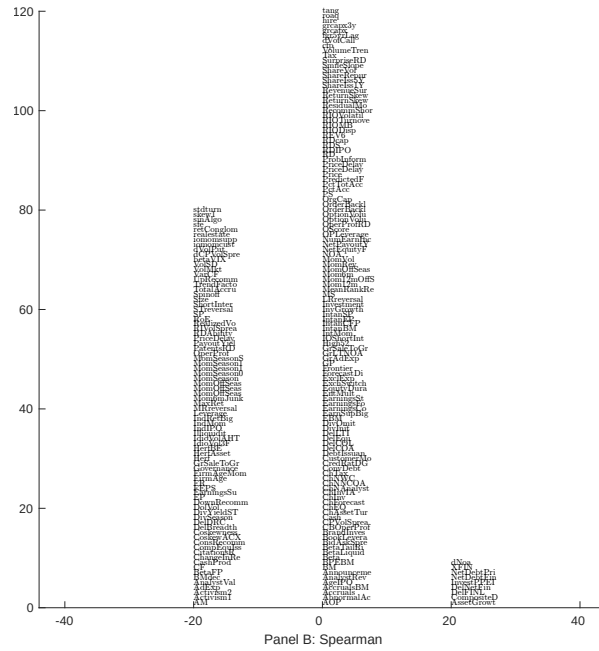
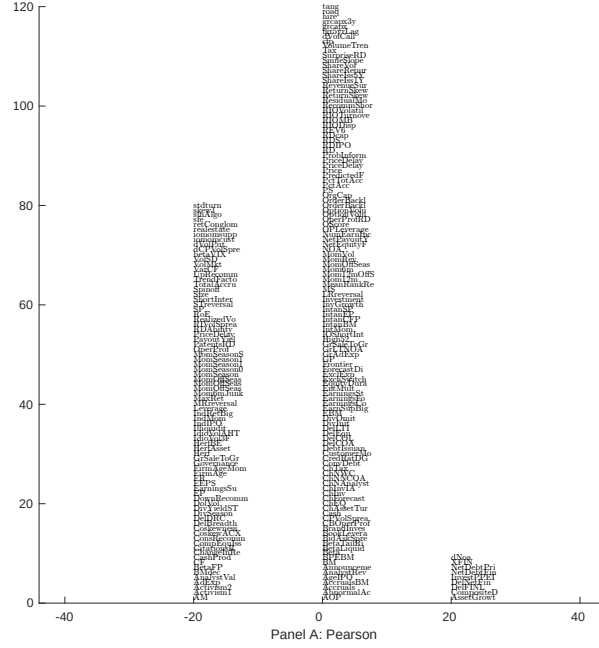


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 209 filtered anomaly signals with IBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

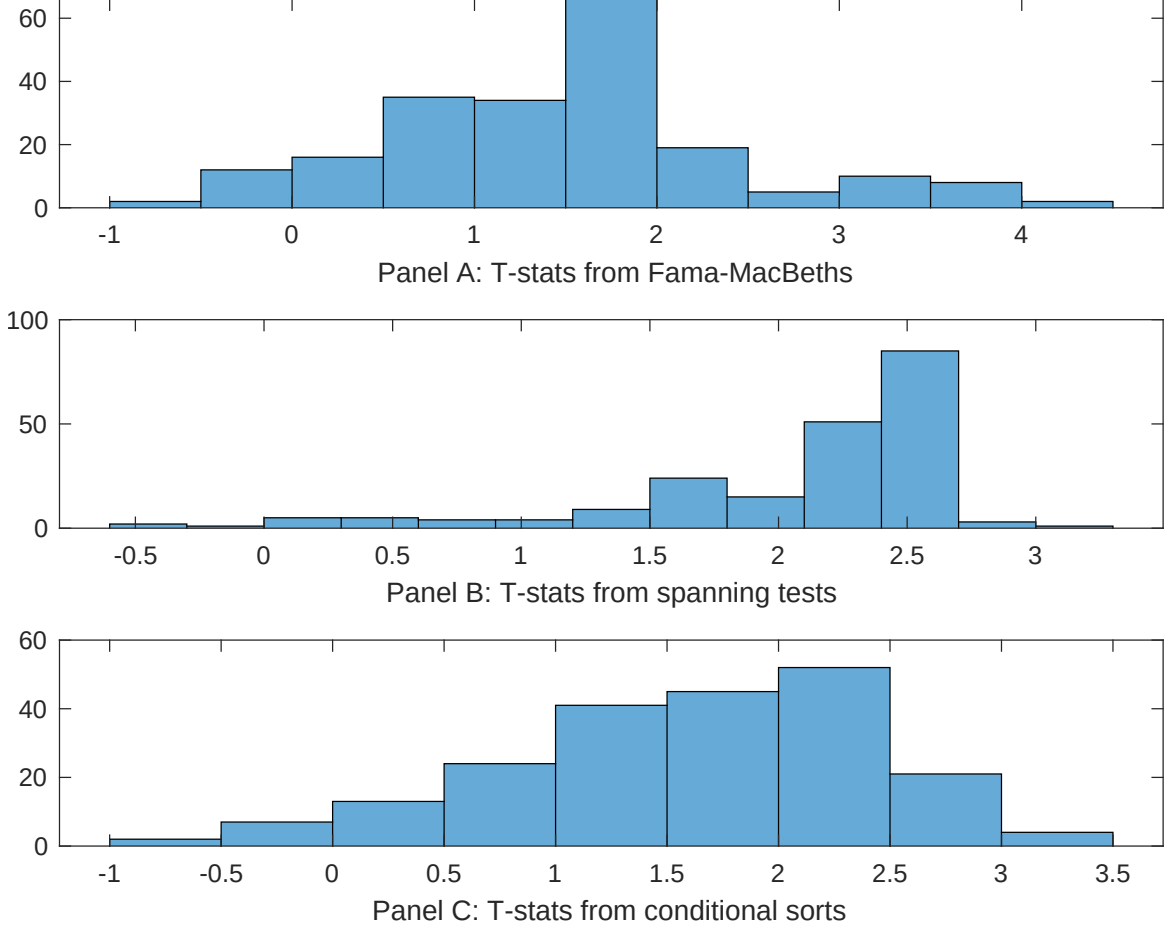


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of IBR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IBR}IBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IBR. Stocks are finally grouped into five IBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IBR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IBR}IBR_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12 [5.67]	0.12 [5.55]	0.13 [5.17]	0.13 [6.31]	0.13 [5.49]	0.10 [4.35]	0.11 [4.50]
IBR	-0.54 [-0.10]	0.19 [0.36]	0.70 [2.72]	0.11 [1.41]	0.61 [2.45]	0.45 [0.55]	-0.31 [-0.10]
Anomaly 1	0.18 [10.31]						0.62 [1.20]
Anomaly 2		0.96 [6.77]					0.30 [0.09]
Anomaly 3			0.20 [9.25]				-0.54 [-0.78]
Anomaly 4				0.95 [5.05]			0.56 [3.18]
Anomaly 5					0.21 [7.23]		0.14 [2.20]
Anomaly 6						0.20 [2.69]	0.13 [2.55]
# months	732	732	630	732	630	732	628
$\bar{R}^2(\%)$	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the IBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IBR} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15 [2.20]	0.14 [1.95]	0.13 [1.72]	0.19 [2.71]	0.13 [1.64]	0.24 [3.44]	0.15 [1.99]
Anomaly 1	35.32 [8.84]						28.63 [4.80]
Anomaly 2		23.07 [6.81]					13.59 [3.40]
Anomaly 3			21.52 [5.03]				-10.76 [-1.92]
Anomaly 4				15.94 [4.71]			5.15 [1.38]
Anomaly 5					10.90 [2.75]		8.93 [2.32]
Anomaly 6						17.54 [5.93]	16.50 [5.15]
mkt	-6.12 [-3.72]	-6.60 [-3.94]	-5.88 [-3.25]	-6.08 [-3.55]	-4.36 [-2.27]	-5.93 [-3.50]	-2.40 [-1.34]
smb	0.03 [0.01]	5.01 [2.06]	1.89 [0.69]	1.22 [0.48]	7.46 [2.46]	1.90 [0.77]	2.54 [0.85]
hml	-9.93 [-3.11]	-12.77 [-3.94]	-12.04 [-3.47]	-14.87 [-4.48]	-11.96 [-3.39]	-15.58 [-4.73]	-13.54 [-4.03]
rmw	5.95 [1.85]	12.00 [3.64]	7.68 [2.16]	7.56 [2.27]	3.67 [0.86]	5.56 [1.66]	0.75 [0.19]
cma	4.43 [0.90]	21.93 [4.57]	11.29 [2.10]	13.73 [2.79]	10.21 [1.73]	4.45 [0.84]	-6.64 [-1.08]
umd	-5.51 [-3.32]	-3.17 [-1.91]	-4.24 [-2.34]	-3.43 [-2.02]	-3.06 [-1.68]	-3.08 [-1.84]	-5.47 [-3.19]
# months	732	732	630	732	630	732	630
$\bar{R}^2(\%)$	14	11	8	8	6	10	20

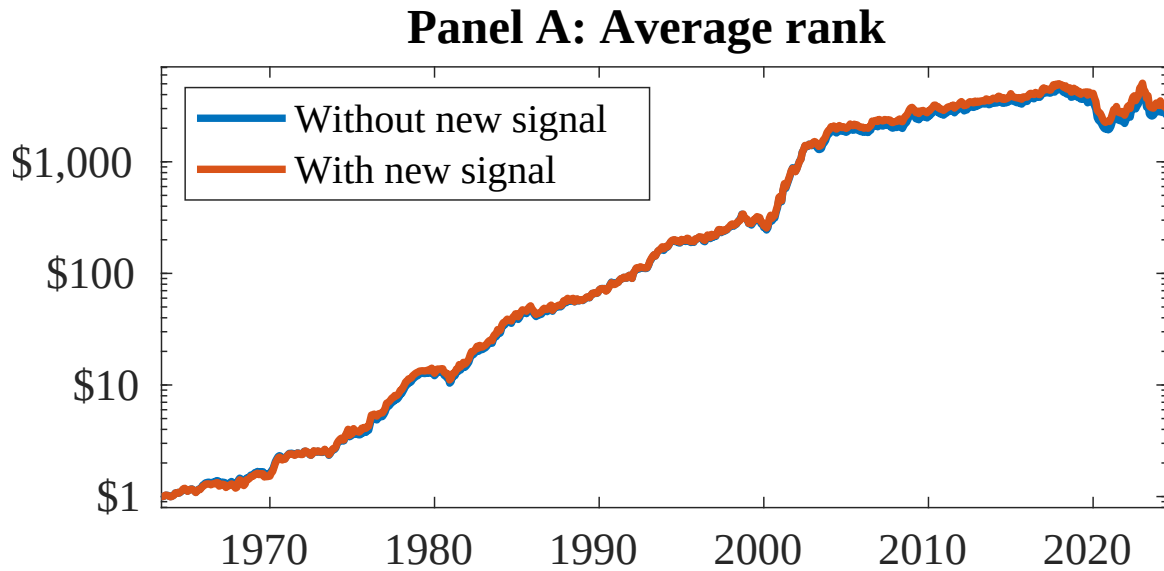


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IBR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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